

What We Did Today

A simple, no-jargon walkthrough of today's conversation - what was asked, and what happened

1 September 2026

Today finished off the small-idea testing from the last session (two more real tests, one a dead end and one a real-but-limited lever), then made a start on properly planning the big remaining idea: letting the model sell extra power abroad, not just buy it.

1. Testing a Real Start-Up Cost for Power Plants

You asked: "Yes, test the cycling cost idea."

Sourced a real number from a genuine published study, built a separate test, and ran it.

Real power plants cost real money to shut down and restart, so they sometimes keep running even at a low price rather than switch off. Our model currently gives every plant a shut-down cost of exactly zero. Found a real, peer-reviewed study on actual German power plant start-up costs and used its real figures (roughly EUR50,000-70,000 per start for a large coal/lignite plant, EUR60,000 for a gas plant, converted into our model's units). Built this as its own separate model version, kept completely apart from every other version, and ran a full simulation.

Result: zero difference, at any point in the whole year.

Every single one of the 8,760 hours of the year came out at EXACTLY the same price, to the cent, with or without the new setting. Investigated why by looking inside AMIRIS's own program: the number IS genuinely calculated somewhere inside it, but that calculation is never actually connected to the price a power plant offers when it bids into the market. So this setting exists in the program but simply isn't wired up to anything that matters here - a cleaner, more conclusive dead end than the price-cap idea from yesterday, since there's no trade-off to weigh at all.

In plain terms: Like flipping a light switch on a wall that was never actually connected to any wire - the switch is real, but nothing happens when you use it.

2. Trying Something That Actually Works: Lignite's Bidding Floor

You asked: "Yes, source a real value and test it."

Checked the mechanism FIRST this time, learning from yesterday's surprise.

Before spending time researching a number, opened up the exact formula our model uses to set a power plant's price - and this time confirmed it really is connected. Our model currently lets lignite (brown coal) plants bid as low as -60 EUR/MWh below their true cost, by far the most generous allowance of any fuel type. There's no published real-world figure for this exact setting, so instead anchored the test to something we could look up: how often Germany's REAL electricity market actually goes negative (3-5% of the time), compared to our model's much higher 18%.

Result: tested four values (-60, -40, -20, -10) - a real effect, but not on the number that matters.

This time 6,218 of the year's 8,760 hours changed price - proof the setting really is connected, a genuine difference from the cycling-cost test. The average price gap to Brainpool and how often we go negative both improved a little as the setting was narrowed - small, real, welcome improvements. But our main trustworthy score for how well the model's DAY-TO-DAY SHAPE matches Brainpool's barely moved at all across every

single value tested. A simpler-looking overall score did jump around a lot, but not in a sensible, step-by-step way - which is exactly the kind of misleading pattern we've caught before in this project, caused by a handful of unusual hours, not real improvement.

Decision: a real, working idea - just not one worth adopting.

Both new tests, and the price-cap test from yesterday, are now fully written up in the Progress Report and added as new slides in the supervisor deck. Between the three, we've now properly tested every reasonably-sized idea available - which makes the case for the one big remaining option (cross-border export) considerably stronger.

3. Starting to Plan the Big One: Selling Power Abroad

You asked: "Let's start scoping the export/market-coupling build."

Found AMIRIS's own real, ready-made template for exactly this, and worked out what a genuine version would actually require.

AMIRIS comes with a working example of two market 'zones' trading with each other in both directions, not just one buying from the other like our current setup. Studied it closely: it needs our existing single market swapped for a two-way version, a small 'coupling' referee agent added to match up both sides, and - the big piece - an entire SECOND country-scale model built alongside ours, with its own power plants, its own renewable energy, and its own demand, plus real hour-by-hour data on how much power can physically flow between the two. Confirmed that real data for the cross-border capacity part does exist and is freely available (the same kind of official European source already used for our import numbers).

In plain terms: Everything we've built so far is one country with a mailbox that can only receive packages (imports). This next step is building a second country next door AND installing a two-way road between them - a much bigger undertaking than any adjustment tried so far, which is exactly why it was saved for last.

This is just the start of the planning - no building has begun yet. The next step is a specific scoping question about how big to make that 'second country': one simplified combined stand-in for the rest of Europe (matching how our import numbers already work), or several real, separately-modelled neighbouring countries (more realistic, but considerably more work). This decision will shape how big an undertaking the whole build turns out to be.